Raffles Institution
Physics Department

Practical WS14: Data-Based Questions **Time: 1 hr**

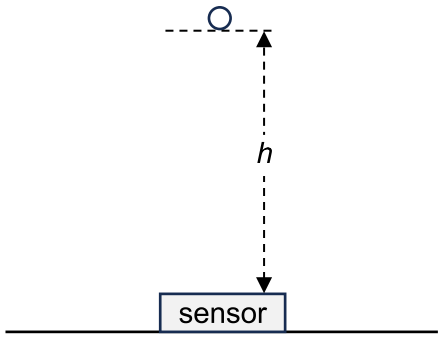
Name: SOLUTIONS ( )

Class: 26S0 **Date:**

The data for this worksheet is found in the respective tabs in the digital file: datasetWS14.xlsx.

Download the file from Teams and submit it in Assignments at the end of the lesson.

1 A ball is released from rest above a sensor that records the variation of the ball's height h with time t as it falls.



(a) **(i)** In column C, calculate t^2.

(ii) Write the formula used in cell C2.

= A2^2 [1]

(b) **(i)** In column D, calculate the displacement s of the ball from its initial position. Take downward displacement as positive.

(ii) Write the formula used in cell D2.

= 1.5–B2 [1]

(c) **(i)** Insert the displacement-time (s–t) scatter chart to detect any anomalous data.

(ii) Identify the 4 values of time t at which anomalous data are observed.

0.06 s, 0.24 s, 0.36 s, 0.52 s [1]

(ii) Delete the 4 rows of anomalous data and the last 8 rows of data when the ball's height is constant as they will skew the trendline.

(d) **(i)** In column E, calculate the velocity v of the ball at each time t using the SLOPE function on the data in columns A and D, except at t = 0 when v = 0. Type "0" in cell E2.

(ii) Write the formula used in cell E3.

= SLOPE(D2:D4,A2:A4) [1]

(e) **(i)** In column F, calculate v^2.

(ii) Write the formula used in cell F2.

= E2^2 [1]

(f) **(i)** Insert the scatter charts for

1. s–t^2

2. v–t

3. v^2–s

(ii) Rename the title of each chart according to part **(f)(i)**.

(iii) Add linear trendline and display the equation of the trendline on each chart.

(iv) Format the trendline label such that the gradient and the vertical-intercept show at least 3 s.f..

(v) Complete the table below for each of the chart. Show your working.

chart	value of gradient to 3 s.f.	calculation of acceleration of free-fall to 3 s.f.
s–t^2	4.96	From $s = ut + \frac{1}{2}at^2$, gradient $= \frac{1}{2}a$ $a = 2 \times$ gradient $= 2 \times 4.96$ $= 9.92$ m s^{-2}
v–t	9.82	From v $= u + at$, $a =$ gradient $= 9.82$ m s^{-2}
v^2–s	19.4	From $v^2 = u^2 + 2as$, gradient $= 2a$ $a = \frac{1}{2} \times$ gradient $= \frac{1}{2} \times 19.4$ $= 9.70$ m s^{-2}

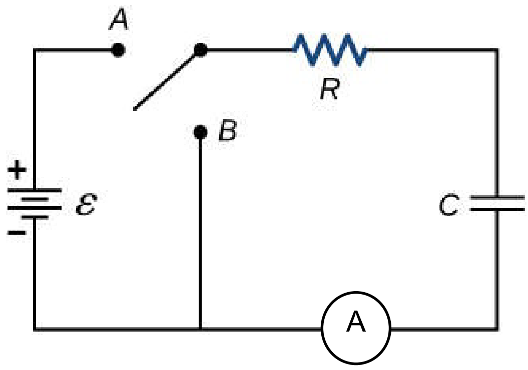
[3]

(vi) Explain which chart provides the most accurate value of the acceleration of free-fall.

The s–t^2 chart provides the most accurate value of a because the data used is least processed.

In addition, the SLOPE function used to determine v gives a poor estimate of the instantaneous gradient at a point on a curve. Hence, the v and v^2 data in inaccurate. [1]

2 A capacitor is fully charged using a battery of e.m.f. ε by closing switch A.



The capacitor is then discharged by closing switch B.

The variation with time t of the discharge current I is given by

$$I = I_0\, e^{-t/\tau},$$

where I_o is the initial current at time t = 0 and τ is the time constant of the circuit.

This equation can be linearised as

$$\ln I = \ln I_0 - \frac{1}{\tau} t .$$

(a) **(i)** Insert the current-time (I–t) scatter chart to detect any anomalous data.

(ii) Identify the 2 values of time t at which anomalous data are observed.

0.60 s, 1.04 s [1]

(iii) Delete the 2 rows of anomalous data.

(b) By applying the trapezium rule in column C to the appropriate section of the I–t graph, determine the amount of charge Q that flows through the discharging circuit from t = 0.64 s to t = 1.12 s.

Q = 0.281 C [1]

(c) **(i)** In column D, calculate ln I.

(ii) Write the formula used in cell D2.

= LN(B2) [1]

(d) **(i)** Insert the scatter chart for ln I–t.

(ii) Rename the title of the chart according to part **(d)(i)**.

(iii) Add linear trendline and display the equation of the trendline on the chart.

(iv) Format the trendline label such that the gradient and the vertical-intercept show at least 3 s.f..

(v) Write the equation of the trendline, with the gradient and the vertical-intercept to 3 s.f..

$\ln I = -2.50t + 1.61$ [1]

(e) Determine the time constant τ of the circuit. Show your working.

$$\text{gradient} = -\frac{1}{\tau}$$

$$\tau = \frac{-1}{\text{gradient}} = \frac{-1}{-2.50} = 0.400 \text{ s}$$

τ = 0.400 s [1]

(f) Given that R = 5.00 Ω, determine, with working,

(i) the e.m.f. ε of the battery, and

$$\text{vertical-intercept} = \ln I_0$$

$$I_0 = e^{1.61} = 5.00 \text{ A}$$

OR

from data file, at t = 0 s, I_0 = 5.000 A.

Hence, e.m.f. $\varepsilon = I_0R = 5.00 \times 5.00 = 25.0$ V.

ε = 25.0 V [1]

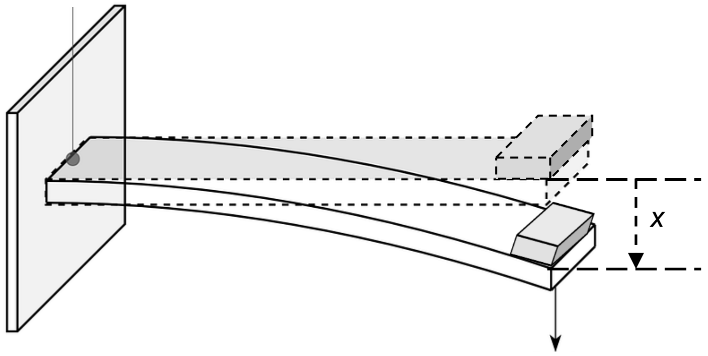
(ii) the capacitance C of the capacitor.

$$\tau = RC$$

$$C = \frac{\tau}{R} = \frac{0.400}{5.00} = 0.0800 \text{ F or } 80.0 \text{ mF}$$

C = 0.0800 F [1]

3 A loaded cantilever is vibrating in simple harmonic motion.



The displacement x of the load from its equilibrium position is related to its acceleration a by

$$a = -\omega^2 x\ ,$$

where ω is the angular frequency.

(a) **(i)** Insert the velocity-time (v–t) scatter chart.

(ii) Identify any anomalous data.

There is no anomalous data. [1]

(b) By applying the trapezium rule in column C to the appropriate section of the v–t graph, determine the amplitude x_o of oscillation.

By studying the v–t graph and the data, it is observed that amplitude x_o can be found from the area under the v–t graph between t = 0 s and t = 0.50 s.

x_o = 0.159 m [1]

(c) **(i)** In column D, calculate the displacement x of the load after each time t using the data in columns A and B, except at t = 0 when x = 0. Type "0" in cell D2.

(ii) Write the formula used in cell D3.

[**Hint:** displacement is measured w.r.t. the equilibrium, hence a cumulative formula is required.]

= 0.5*(A3-A2)*(B3+B2) + D2 [1]

(d) **(i)** In column E, calculate the acceleration a of the load at each time t using the SLOPE function on the data in columns A and B, except at t = 0 when a = 0. Type "0" in cell E2.

(ii) Write the formula used in cell E3.

= SLOPE(B2:B4,A2:A4) [1]

(e) **(i)** Insert the scatter chart for a–x.

(ii) Rename the title of the chart according to part **(e)(i)**.

(iii) Add linear trendline and display the equation of the trendline on the chart.

(iv) Format the trendline label such that the gradient and the vertical-intercept show at least 3 s.f..

(v) Write the equation of the trendline, with the gradient and the vertical-intercept to 3 s.f..

$a = -9.87x - 0.00128$ .... [1]

(f) Determine the value of ω. Show your working.

$$\text{gradient} = -\omega^2$$
$$\omega = \sqrt{-\text{gradient}}$$
$$= \sqrt{-(-9.87)}$$
$$= 3.14 \text{ rad s}^{-1}$$

ω = 3.14 rad s^{-1} .... [1]

(g) **(i)** Insert the scatter charts for x–t and a–t.

(ii) Comment on your observations of the two charts.

Any valid comment related to the relationship $a \propto -x$.

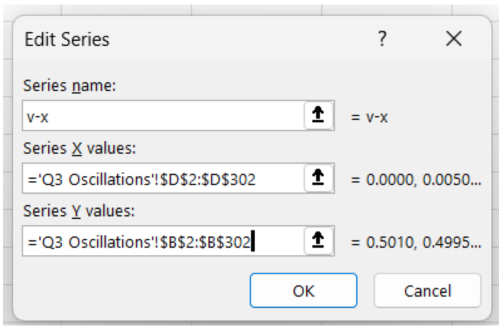
E.g.: "x-t is a sine curve, while a-t is a negative sine curve." .... [1]

(h) **(i)** Insert the scatter chart for v–x.

Note: As the v column is on the left and the x column is on the right, Excel will plot the x-v curve instead.

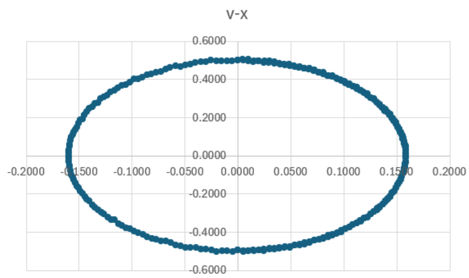
To change it to the v-x curve, you will need swap the axes being plotted by doing the following:

1. Right-click on any data point and choose "Select Data…" on the pop-up dialogue box.
2. In the next dialogue box, click "Edit" to rename the Series as "v-x" and reselect the correct X values in column D and Y values in column B as shown.



Alternatively, you can copy-and-paste the v column to the right of the x column.

(ii) Sketch the graph in the space below.



[1]

End of Paper